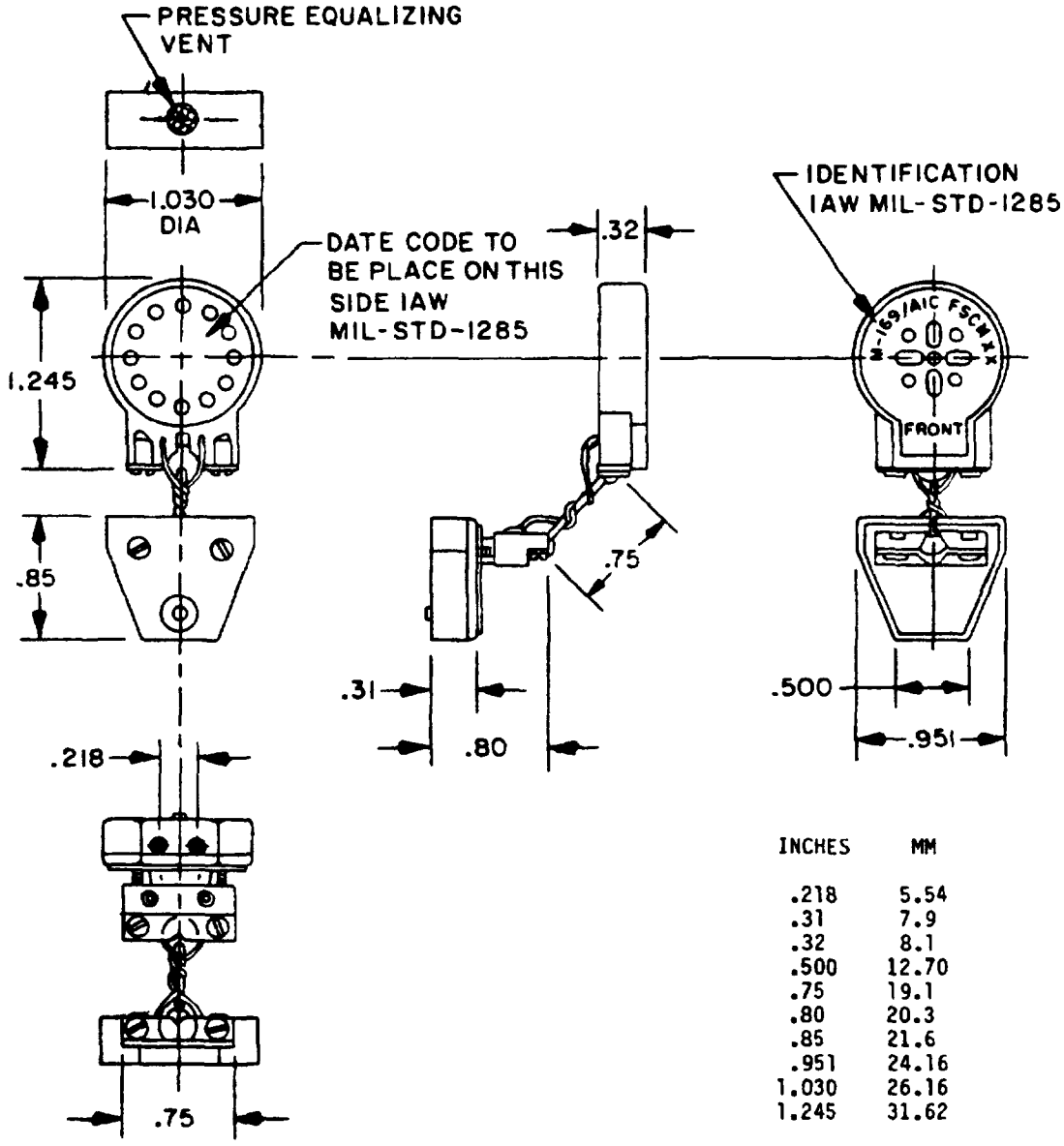
MILITARY SPECIFICATION SHEET

MICROPHONE, M-169/AIC

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the microphone described herein shall consist of this specification and the latest issue of MIL-M-26542.



INCHES	MM
.218	5.54
.31	7.9
.32	8.1
.500	12.70
.75	19.1
.80	20.3
.85	21.6
.951	24.16
1.030	26.16
1.245	31.62

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.015 (0.38 mm).

FIGURE 1. Microphone element M-169/AIC.

FSC 5965

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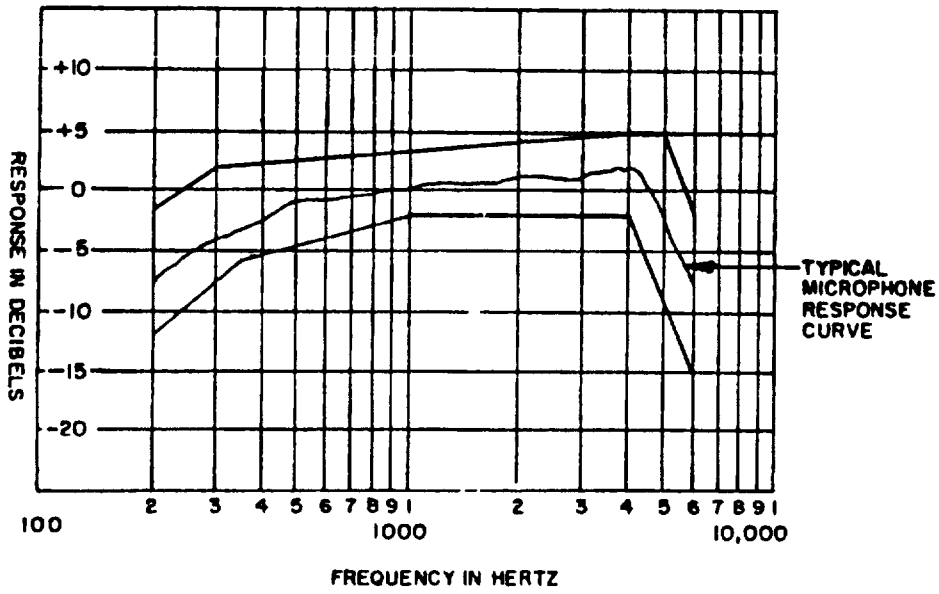


FIGURE 2. Frequency response at sea level.

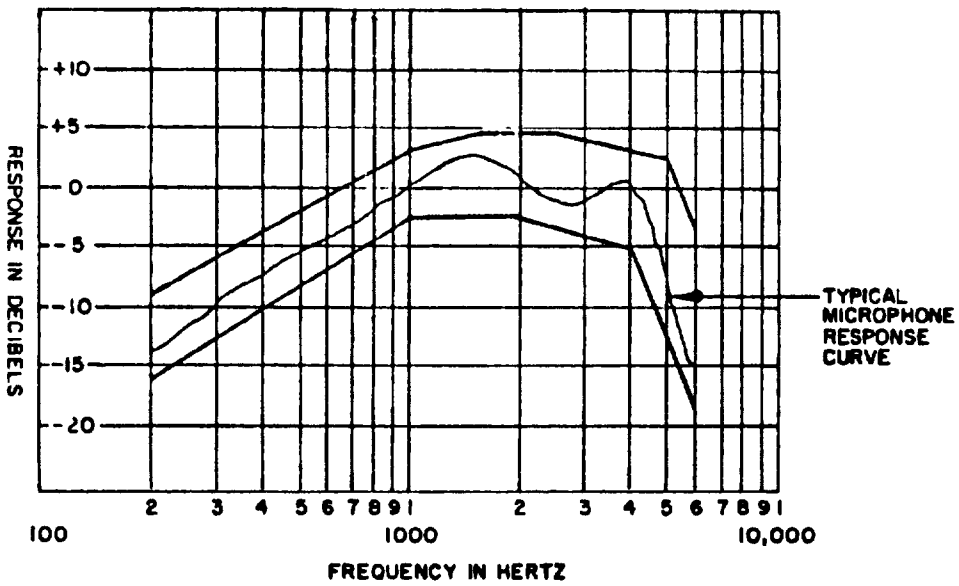


FIGURE 3. Frequency response at 25,000 feet.

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REQUIREMENTS:

Moisture barrier seal test: The microphone element shall be immersed in water and pressure of plus one-quarter pound per square inch shall be applied to the pressure-equalization hole to determine whether there are any leaks through the microphone element and around the moisture barrier seal.

Reliability: The reliability requirements of MIL-R-25717 shall apply as requirements of this specification with the following additions and exceptions:

a. Mean-Time-Between-Failures (MTBF) Requirements. The M-169/AIC microphone element shall provide an MTBF of 500 hours during service life.
b. Longevity (Normal Operating Period). The M-169/AIC microphone element shall have a longevity or minimum normal operating period of 3,000 hours before wearout failures occur or the MTBF becomes less than the value specified.
c. Aging. The microphone element shall not contain parts fabricated from materials known to changes their properties appreciably because of aging.

Color: Semigloss black, 27038 in accordance with FED-STD-595.

Component parts: The microphone shall be supplied with the microphone element M-169/AIC (see figure 1).

Weight: The weight of the microphone element shall not exceed 16 grams.

Performance:

Sensitivity at ground level: 2.5 to 5.0 microvolts referred to 74 dB relative to 20 micropascals.

Sensitivity at a simulated altitude of 25,000 feet: Equal to but not more than 9 dB of the ground level sensitivity.

Frequency response at ground level and at 25,000 feet: The envelope shall be as shown on figures 2 and 3. The frequency response range of the element shall be 200-6,000 Hz.

Impedance. 4-6 ohms. The electrical impedance at any frequency over the range of 200 to 6,000 Hz shall not deviate from the 1,000 Hz impedance by more than 20 percent.

Resistive load: 5 ohms.

Intended use: Microphone element M-169/AIC is a noise-canceling dynamic moving coil microphone element designed for use in an oxygen mask or pressure-type oxygen helmet, at altitudes where the use of an oxygen helmet is required. The microphone element is intended for use with intercommunication sets AN/AIC-10( ) and AN/AIC-18 to provide communication under the noise conditions encountered in military aircraft. This microphone's primary intended use is with the MBU-12 Oxygen Mask.

The microphone assembly shall be tested in accordance with the tests listed in table I.

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TABLE I. Parameter applicability.

Inspection	Requirements paragraph	Test paragraph	Qualification tests	Group "A" tests	Group "B" tests	Group "C" tests
Group I						
Visual and mechanical inspection	3.1,3.4,3.5	4.7.1	X	X		
Sensitivity of ground level	3.5.1	4.7.2.2	X	X		
Sensitivity at altitude	3.5.2	4.7.2.3	X			
Frequency response at ground level	3.5.3	4.7.2.4	X	X		
Frequency response at altitude	3.5.4	4.7.2.5	X			
Impedance	3.5.5	4.7.3	X	X		
Noise cancellation characteristics	3.5.7.1	3.7.5.1	X			
Effect of external magnetic field	3.5.8	4.7.6	X			
Stray magnetic field	3.5.9	4.7.7	X			
Linearity	3.5.10	4.7.8	X			
Talk out	3.5.11	4.7.9	X	X		
Dielectric withstanding voltage	3.5.12	4.7.10	X			
Signal to noise	3.5.7	4.7.5	X		X	
Distortion	3.5.6	4.7.4	X		X	
Interchangeability	3.5.13	4.7.11	X		X	
Group II						
Thermal shock	3.5.14	4.7.12	X			X
Humidity	3.5.15	4.7.13	X			X
Drop	3.5.16	4.7.14	X			X
Pressure equalization	3.5.17	4.7.15	X			X
Explosive decompression	3.5.18	4.7.16	X			X
Salt fog	3.5.19	4.7.17	X			X
Group III						
Vibration	3.5.20	4.7.18	X			X
Bounce	3.5.21	4.7.19	X			X
Altitude	3.5.22	4.7.20	X			X
Moisture barrier	3.5.23	4.7.21	X			X
Immersion	3.5.24	4.7.22	NA			NA
Group IV						
Fungus	3.5.25	4.7.23	NA			
Group V						
Gun blast	3.5.26	4.7.24	NA			

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - AS
Air Force - 99
DLA - ES

User activities:
Army - AR, AT, AV, ME
Navy - OS

Agent:
DLA - ES

Preparing activity:
Air Force - 85

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